

Dundee neutrophil isolation protocol (from whole blood)

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1. MATERIALS

1.1. Consumables:

- 1.1.2. BD Vacutainer (BD Vacutainer™ Hemogard Closure Plastic K2-EDTA Tube, Cat# BD 367525)
- 1.1.3. 50 mL falcon tube (CELLSTAR, Cat #227 261).
- 1.1.4. 15 mL falcon tube (CELLSTAR, Cat #188 271).
- 1.1.5. 1.5 mL Eppendorf tubes.
- 1.1.6. Marker pen.
- 1.1.7. Ice.
- 1.1.8. Liquid nitrogen.
- 1.1.9. Dry Ice and large polystyrene box for shipping
- 1.1.10. Ethanol, in spray bottle.
- 1.1.11. Pipette tips (serological 10 mL, 25 mL & 1 mL, 200 µL, 10 µL).
- 1.1.12. Personal protection equipment:
 - 1.1.12.1. Disposable gloves.
 - 1.1.12.2. Lab coat.
 - 1.1.12.3. Safety glasses.

1.2. Reagents:

- 1.2.1. EasySep™ Direct Human Neutrophil Isolation Kit (Stemcell technologies, Catalog #19666). This contains Solutions called “Isolation Cocktail” and “RapidSpheres magnetic beads”.

- 1.2.2. Dulbecco's phosphate-buffered saline (PBS) (ThermoFisher Cat# 14190094).
- 1.2.3. RPMI 1640 Medium (ThermoFisher Cat# 21875034).
- 1.2.4. MLi-2 (LRRK2 inhibitor diluted in DMSO at a stock concentration of 200 μ M (1000X concentration), available from MRC-PPU Reagents; to order please email: MRCPPUreagents@dundee.ac.uk and address request to both Hilary McLauchlan and James Hastie
- 1.2.5. DMSO (Sigma, Cat #D8418).
- 1.2.6. Isopropanol (anhydrous grade) (Sigma, Cat #278475)
- 1.2.7. DIFP (Diisopropylfluorophosphate) (Sigma, Cat #D0879). Prepare 0.5M stock solution in isopropanol using special precautions (please see 1.2.9.)
- 1.2.8. Prepare two stocks of EDTA solution:
 - 1.2.8.1. Prepare 0.1 ml of EDTA Stock Solution 1 containing 100 mM Ethylenediaminetetraacetic acid (EDTA) (Sigma, Cat #E6758) in phosphate-buffered saline (PBS) (ThermoFisher Cat# 14190094). This is called (EDTA stock solution 1). 100 μ l of EDTA stock solution 1 is added to each 10 ml of human blood (Step 2.3). This solution is made by dissolving EDTA powder in PBS and then adjusting pH to 8.0 to allow EDTA to dissolve. This solution is available to order please email: MRCPPUreagents@dundee.ac.uk and address request to both Hilary McLauchlan and James Hastie.
 - 1.2.8.2. Prepare 60 ml of EDTA stock solution 2 containing 1 mM Ethylenediaminetetraacetic acid (EDTA) (Sigma, Cat #E6758) in Dulbecco's phosphate-buffered saline (PBS) (ThermoFisher Cat# 14190094). This is called (EDTA stock solution 2). About 40 ml of EDTA stock solution 2 needed for Step 2.8 in the protocol and approximately 6-11 ml is required for Step 2.19.
- 1.2.9. Lysis buffer (50 mM Tris-HCl pH 7.5, 1%(v/v) Triton X-100, 1 mM EGTA, 1 mM Na₃VO₄, 50 mM NaF, 10 mM β -glycerophosphate, 5 mM sodium pyrophosphate, 0.27 M sucrose, 0.1% (v/v) β -mercaptoethanol, 1x cOmplete(EDTA-free) protease inhibitor cocktail (Roche), 1 μ g/ml Microcystin-LR, 0.5 mM diisopropylfluorophosphate (DIFP).

- 1.2.9.1. Note: Diisopropylfluorophosphate (DIFP) is prepared as a 0.5 M stock solution in isopropanol (it is unstable in aqueous solution) and stored at -80 °C. Please note that DIFP is toxic and should be handled with care in the fume hood. DIFP can be added to the lysis buffer and used immediately. Alternatively, complete lysis buffer containing DIFP and all other components can be aliquoted and stored at -80 °C for subsequent use for at least 4 weeks.
- 1.2.9.2. Note we purchase Microcystin-LR from Enzo Life Sciences, Cat# number ALX-350-012-M001) and make 1 mg/ml stock in DMSO and store at -80 °C.
- 1.2.9.3. We can provide frozen lysis buffer in aliquots without Microcystin-LR and DIFP; to order please email MRCPPUreagents@dundee.ac.uk and address request to both Hilary McLauchlan and James Hastie.

1.3. Equipment:

- 1.3.1. Category 2 biological safety cabinet.
- 1.3.2. Easy 50 EasySep™ Magnet (Stemcell technologies, Catalog #18002).
- 1.3.3. Centrifuges (Beckman Coulter Allegra X-15R & Eppendorf centrifuge 5417R) or equivalent. Swinging Bucket Rotor for 15 mL and 50 mL falcon tubes at speed of 1000-1200 xg.
- 1.3.4. Pipette tips and pipettes.
- 1.3.5. Freezer, -80 °C.
- 1.3.6. Liquid nitrogen carrier.

2. METHOD

According to local UK regulation we undertake all manipulations and pipetting of human blood in a category 2 biological safety cabinet.

- 2.1. Collect 10 ml of blood into BD Vacutainer (BD Vacutainer™ Hemogard Closure Plastic K2-EDTA Tube, Cat# BD 367525). Ensure 10 ml of blood is collected by not prematurely discontinuing blood collection. Mix gently by inverting tubes 7-8 times.

- 2.2. Transfer 10 mL of blood into a 50 mL falcon tube.
- 2.3. Add to blood 100 μ l of “EDTA Stock Solution 1” (0.1M EDTA-PBS solution). Mix gently.
- 2.4. Add 0.5 ml “isolation cocktail solution” from the neutrophil isolation kit to the whole blood sample.
- 2.5. Vortex the “RapidSpheres magnetic beads” from the neutrophil isolation kit for 30 secs before use in order to resuspend very fine magnetic beads.
- 2.6. Add 0.5 ml RapidSpheres magnetic beads to the blood sample.
- 2.7. Mix blood sample gently by inverting the tube and incubate at room temperature for 5 mins.
- 2.8. Top tube up to 50 ml with “EDTA Stock Solution 2” (1 mM EDTA-PBS solution).
- 2.9. Mix by very gently pipetting up and down 2-3 times.
- 2.10. Place the tube into the magnet and remove the lid to avoid subsequent agitation of tube. Incubate for 10 minutes at room temperature.
- 2.11. Carefully pipette the enriched cell suspension that contains neutrophils into a new 50 ml falcon tube. Take care not to touch the side of the tube that is in contact with the magnet and avoid collection and perturbation of the red blood cells at the bottom of the tube (leave approximately 10 ml of the red blood cell suspension behind at the bottom of the tube).
- 2.12. Add 0.5 mL RapidSpheres magnetic beads to the tube from Step 2.11.
- 2.13. Mix blood sample gently by inverting the tube and incubate at room temperature for 5 mins.
- 2.14. Place the tube into the magnet, remove the lid, and leave for 5 mins at room temperature.
- 2.15. Carefully pipette the enriched cell suspension that contains neutrophils into a new 50 ml falcon tube. Take care not to touch the side of the tube that is in contact with the magnet. Leave approximately 5ml of the suspension at the bottom of the tube.
- 2.16. To ensure complete removal of magnetic beads from cell mixture, place the new tube from Step 2.15 into the magnet, remove the lid, and leave for 10 min at room temperature.
- 2.17. Carefully pipette the enriched cell suspension that now contains pure neutrophils into a new 50 ml falcon tube. Take care not to touch the side of

the tube that is in contact with the magnet. Leave approximately 5 ml of the suspension at the bottom of the tube.

- 2.18. The neutrophils will be in a volume of 30-35 ml at this stage.
- 2.19. Top up isolated cells with 1 mM EDTA stock solution 2 to a final volume of approximately 41 ml. Pipette up and down to mix. Divide solution equally into 2 tubes with approximately 20 ml in each tube.
- 2.20. Centrifuge both tubes at 335 x g for 5 mins. The setting for acceleration and deceleration is both 5 using a Beckman Coulter Allegra X-15R Centrifuge (maximum is 10).
- 2.21. During this centrifugation take MLI-2 inhibitor stock (200 μ M / 1000X concentration) out of the -80 freezer, leave at room temperature for subsequent use (Steps 2.26 and 2.28).
- 2.22. Immediately after the centrifugation, carefully discard the supernatant without disturbing the neutrophil pellet.
- 2.23. Resuspended each cell pellet in 10 ml of RPMI medium at room temperature by gently pipetting cells up and down 4 times.
- 2.24. Label one tube "DMSO" and other tube "MLi-2".
- 2.25. To "DMSO" labelled tube, add 10 μ l of DMSO and mix gently by pipetting up and down 2-times with a 10 ml pipette.
- 2.26. To "MLi-2" labelled tube, add 10 μ l of 200 μ M MLI-2 stock solution (final concentration 200 nM) and mix gently by pipetting up and down 2-times with a 10 ml pipette.
- 2.27. Incubate samples for 30 minutes at room temperature. Mix gently by inversion every 10 minutes during the incubation.
- 2.28. During the incubation period,
 - remove 0.5 M DIFP stock from -80 °C freezer and place in fume hood on ice.
 - remove 1 mg/ml Microcystin-LR stock from -80 °C freezer and at room temperature to thaw.
 - defrost an aliquot (0.25ml) of the lysis buffer by taking it out of the freezer, allow to defrost at room temperature and then place it on ice for subsequent use (Step 2.34).
 - prepare 1 ml of RPMI medium containing 1 μ l of DMSO and call this "DMSO Resuspension Buffer".

- prepare 1 ml of RPMI medium containing 1 μ l of 200 μ M MLi-2 and call this “MLi-2 Resuspension Buffer”.
- 2.29. After the 30 min incubation, centrifuge both tubes at 335 x g for 5 mins. The setting for acceleration and deceleration is both 5 using a Beckman Coulter Allegra X-15R Centrifuge (maximum is 10) to pellet the neutrophils.
 - 2.30. Carefully discard the supernatant in each tube without disturbing the neutrophil pellet.
 - 2.31. For the DMSO labelled sample gently resuspend pellet in 1 ml of “DMSO resuspension buffer” and transfer to Eppendorf tube labelled “DMSO”.
 - 2.32. For the MLi-2 labelled sample gently resuspend pellet in 1 ml of “MLi-2 resuspension buffer” and transfer to Eppendorf tube labelled “MLi-2”.
 - 2.33. Centrifuge both tubes at 300 x g for 3 min.
 - 2.34. During this time proceed with preparing the lysis buffer: In the fume hood carefully to the 0.25 ml lysis buffer add 0.25 μ l of 0.5 M DIFP solution as well as 0.25 μ l of 1 mg/ml microcystin-LR. Mix and leave on ice until use. DIFP should be added to lysis buffer within 15 min of cell lysis (Step 2.38 and 2.39) as DIFP relatively unstable in aqueous solution.
 - 2.35. Return 0.5M DIFP stock to -80 °C freezer at this stage.
 - 2.36. After the centrifugation in Step 2.33, immediately proceed to carefully remove all supernatant with a pipette without disturbing the neutrophil pellets.
 - 2.37. Place the tubes with the neutrophil cell pellets on ice.
 - 2.38. Immediately add 100 μ L of lysis buffer containing DIFP to the “DMSO” labelled tube. Using a 100-200 μ L pipette, re-suspend the cell pellet by pipetting up and down until all cells are lysed (5-10 times).
 - 2.39. Next add 100 μ L of lysis buffer containing DIFP to the “MLi-2” labelled tube. Using a 100-200 μ L pipette and using a 100-200 μ L pipette, re-suspend the cell pellet by pipetting up and down until all cells are lysed (5-10 times).
 - 2.40. Incubate cell lysates on ice for 10 mins. Place “DMSO” and “MLi-2” labelled tubes into a centrifuge and remove cell debris by centrifugation at 14,000 rpm for 15 mins at 4 °C.
 - 2.41. Transfer ”DMSO” and “MLi-2” supernatants (containing neutrophil lysates) into new Eppendorf tubes. Discard debris pellet.
 - 2.42. Immediately snap freeze samples in liquid nitrogen.

2.43. Store samples at -80 °C until further use or shipment on dry ice keeping samples frozen at all times.